QPCODE: **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIFTH SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION

Fifth Semester
MG5DSECFS301- INTERNET OF THINGS
(2024 ADMISSION ONWARDS)

Duration: 2Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any 10 questions

Each question carries **2** marks

1. Explain the meaning of IOT. [U] / [1]

2. Classify the main components in the physical design of IoT. [U] / [1]

3. Describe any two IoT enabling technologies. [U] / [1]

4. Differentiate between physical design and logical design of IoT. [U] / [1]

5. List different layers of IoT networking. [K] / [1]

6. Examine the role of communication protocols in IoT. [An] / [2]

7.	Differentiate between M2M communication and IoT.	[An] / [2]
8.	Compare NETCONF and YANG to state their role in IoT system management.	[An] / [3]
9.	Evaluate the importance of robustness in an IoT system.	[E] / [4]
10.	Select any two factors that can affect the reliability of an IoT system.	[E] / [4]
11.	Evaluate the significance of addressing reliability in IoT-based applications.	[E] / [4]
12.	Select a suitable IoT management tool between Chef and Puppet for automating configuration management, and state one reason for your selection.	[E] / [4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any 6 questions

Each question carries **5** marks

13.	Explain the different layers of IoT architecture with their functions.	[U] / [1]
14.	Describe the major aspects of IoT data management and explain its importance in IoT applications.	[U] / [1]
15.	Illustrate the different layers of IoT networking with suitable examples.	[An] / [2]
16.	Illustrate the characteristics of C and its flavours used in embedded IoT programming.	[An] / [3]
17.	Differentiate nes C, Keil C, Dynamic C and B# based on their characteristics and applications in embedded programming.	[An] / [3]
18.	Combine NETCONF and YANG and illustrate how they support IoT system management.	[An] / [3]

19. Evaluate the major reliability issues in IoT systems and explain their impact on the performance of IoT applications. [E] / [4]

20. Evaluate the working of home automation system. [E] / [4]

[5x6 = 30]

Part C

Essay Questions

Answer any 2 questions

Each question carries **10** marks

21. Classify the major IoT enabling technologies and describe their role in developing IoT applications. [U] / [1]

22. Illustrate IoT networking and system management in detail. [An] / [2]

23. Evaluate the robustness and reliability requirements of IoT systems. [E] / [4]

[10x2 = 20]